CATERPILLAR®

DEALER BEST PRACTICE SERIES

Fluid Cooler Testing Basin

Application | Maintenance | Site Management | **Component Rebuild** | Safety | MARC Management

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1.0 Introduction

Efficient/accurate cooler testing is an important part of the component repair/rebuild process. A fluid cooler that leaks, contains trapped debris, or is plugged can cause component failures. Cooler testing is usually performed with adapters stored in boxes, on open shelves, under workbenches, or randomly hooked on open pegboards. These adapter storage practices can lead to:

- Dirty adapters that can contaminate the system or require cleaning before use.
- Damaged adapters
- Lost or misplaced adapters

This Best Practice describes the test equipment and implementation. A well-designed test area and tooling can improve process efficiency and quality.

Fluid cooler testing capability provides revenue by allowing the dealer to perform testing rather than outsourcing cooler service. It also reduces cooler-related warranty claims.

2.0 Best Practice Description

Using specially designed basin for cooler testing reduces the possibility of contamination and damage by using a special cooler testing basin in a cleaning bay, not in the test and/or group assembly area. Additionally, a dealer should keep all housings, fixtures, pipes, etc., with the cooler group in the bay to reduce chances of contamination.

The basin should be:

- Sized for the dealer's common products.
- Insulated/heated if the dealer chooses to test with heat
- Smooth painted exterior designed with few areas to catch dust/dirt/crime for fast/easy cleaning and professional image.
- Equipped with aluminum, counterbalanced lid for easy operation.

Additionally, the dealer may wish to include:

- An adapter storage drawer built into the basin lower section
    - Reduces space consumption by storing tools in basin footprint
    - Reduces lost adapters by keeping them in securable drawer(s) close to where they are used.
    - Provides easy adapter inventory review by organizing adapters in foam cutouts.
    - Test notes, special instructions, or other references may also be stored in sections of the drawer(s).

- Air regulator manifold permanently mounted and plumbed into the facility air system for ease-of-use (requires only one hand to set/adjust).
    - Manifold may be permanently mounted on test basin.
    - Manifold may be pin-mounted (stored in drawer between uses) for controlled book-outs to other shop/field service departments.

3.0 Implementation Steps

1. **Determine area in shop to be designated for cooler testing.**
    a. Assure test area has supporting utilities (compressed air and electricity as needed)
    b. Assure area has adequate space and access.

2. **Design and construct test basin**

3. **Establish tooling needs and storage requirements.**
    a. Research current tool inventories and test references/notes in dealer.
    b. Research Caterpillar tool recommendations and test references.
    c. Design drawers to include:
        i. Foam cutouts for current adapters with plenty of room for future additions.
        ii. Test references/notes in a shop-consistent format/organization for easy maintenance.

4. **Determine if heated basin water is desired (this is not a Caterpillar test requirement, but is desired by many dealers).**
    a. Choose heater and design into basin for easy maintenance.
    b. Assure designated test area has supporting utilities for heater.

5. **Determine most applicable air-regulator manifold application:**
    a. Design manifold for efficient set-up, adjustment, and storage as needed.
    b. Design basin drawers for portable manifold storage if needed.

6. **Develop/adjust and document test procedure with quality checklists as needed.**

7. **Establish new/adjusted repair/rebuild time requirement targets as needed.**
    a. Eliminate the current processes/tooling/equipment strategy.
    b. Remove replaced items (tools/adapters) from operations to prevent risky applications.
    c. Include inter-department use issues as needed.

8. **Install the layout/equipment changes into the shop**
    a. This may occur in phases, as needed, such as shop arrangement, then equipment final design/fabrication.
    b. Promote change feedback as installation may surface new problems.

9. **Train employees to use the new procedures/equipment.**
    a. Establish current testing concerns with team.
    b. Establish process, layout, and equipment improvements through team.
    c. Operation training, safety training.

10. **Establish equipment maintenance schedule/responsibility.**

11. **Review process/equipment performance once established.**

12. **Establish and install any adjustments as needed.**

4.0 Benefits

- Testing capability remains in-house.
- Labor efficiency gain:
  - Well designed/organized test equipment reduces set-up and teardown time.
  - Integrated/timed (smart) product flow simplifies process and reduces administrative/rebuild labor.
  - Product staged with like-product reduces space requirements.
  - Test delays avoided by keeping test parts/ adapters organized.
- Increases quality:
  - Avoids unexpected cooler failure upon reinstallation to machine.

5.0 Resources Required

Facility

- Re-organize workstation to allow for basin placement, incoming and outgoing staging, and a workbench as needed.

Support Equipment

- Well-built basins with heaters
- A generic work bench to install/remove adapters, and assemble cooler groups as needed.
- Caterpillar test adapters. The dealer may already have these.
- Drawer foam inserts are available Caterpillar and must be custom cut per each shops needs.
- Lifting devices such as cranes may be shared with other functions in the shop. Lift requirements are relatively low for a typical Caterpillar repair/rebuild facility and would be determined by the size of coolers and or cooler groups being handled.
- Lifting chains, straps, and/or brackets may also be shared with other functions.

People

- One trained operator (based on demand).

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

Also see Best Practice #1006-4.4-1023, *Floor Mounted Cooler Assembly Fixtures*.


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Floor Mounted Cooler Assembly Fixtures

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300 Hamilton Blvd., Ste. 300, Peoria, IL 61629-3810, U.S.A. catminer.com

8.0 Acknowledgements

- This Best Practice was written by:

  Russ Young
  Black Belt, Cat Global Mining